<u>Report</u>

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Introduction

This report analyzes the Haunted Places dataset using large-scale content extraction techniques to uncover new insights into paranormal reports across the U.S. We integrated geolocation data with GeoTopicParser, extracted named entities using SpaCY, and generated AI-based visuals for each location. These images were captioned with Tika's Show and Tell model and analyzed using InceptionV3 and ChatGPT Vision to detect objects and interpret scenes. This report presents key findings, including geographic patterns, entity associations, and the impact of visual metadata on understanding the haunted narratives.

Object Detection

We performed object detection on AI-generated haunted place images to analyze the relationship between visual media and textual descriptions. Although the assignment instructed us to use Tika's Inception REST service via Docker, we encountered compatibility issues on Apple Silicon hardware, including platform conflicts and the container defaulting to an unrelated service. As a result, we implemented an equivalent solution locally using TensorFlow's pre-trained InceptionV3 model, which is the same model used in the Tika service.

Our Python script mapped each haunted place's unique ID to its corresponding AI-generated image and used the InceptionV3 model to extract the top five objects in each image. The detected objects were added to our dataset's new column, detected_objects. For example, one model output included: 1. church: 98.69%, 2. lemon: 0.09%, 3. monastery: 0.06%, 4. orange: 0.05%, 5. wild_boar: 0.03%. This result reflects the expected behavior of general-purpose object detection models like InceptionV3. The Tika and TensorFlow implementations of InceptionV3 are trained on the ImageNet dataset, which contains approximately 1,000 labeled categories of everyday objects such as animals, buildings, and food. Although the model confidently predicted everyday objects, the detected labels often had no connection to haunted places or the generated captions, as ImageNet does not include domain-specific categories like "haunted house," "ghost," or other supernatural concepts.

We also observed specific trends in the types of objects recognized by the model. Many detected objects were structural or architectural elements, including "castle," "church," "monastery," and "palace," which likely reflect visual patterns in the AI-generated images associated with haunted locations. However, the model also frequently predicted unrelated or nonsensical objects such as "comic book," "fire screen," or "doormat," underscoring the limitations of applying general-purpose image classification models to stylized, AI-generated content.

In addition to implementing InceptionV3, we also experimented with ChatGPT's Vision capabilities as a complementary method of interpreting the AI-generated haunted images. Unlike traditional object detection models trained on fixed label sets like ImageNet, ChatGPT Vision uses a multimodal transformer architecture that combines image understanding with contextual language reasoning. This allows it to generate rich, human-like captions and infer meaning from surreal or stylized content, an advantage when working with images that intentionally blur realism and fantasy.

For example, one image depicting a gothic building with clock towers and crosses was classified by InceptionV3 as containing a "church" (98.69%) and "monastery" (0.06%), along with low-confidence false positives like "lemon" and "wild_boar." While these results reflect surface-level object recognition, ChatGPT Vision was able to provide a much more accurate and semantically relevant description: "Gothic-style church building with arched windows, a clock tower, gravestones, and a full moon in the background, suggesting a haunted or eerie setting." This contextual interpretation better aligns with the image's visual tone and haunted narrative.

In another case, InceptionV3 identified a scene dominated by chains and dark textures as a "fire screen" and "comic book," whereas ChatGPT Vision recognized it as "a prison-like interior with kneeling, humanoid figures bound by chains, evoking imagery of captivity or suffering." This description offered a more nuanced reading, capturing not only the objects but their emotional and thematic implications, something the Inception model was not designed to do.

These comparisons demonstrate that while InceptionV3 effectively identifies tangible objects, ChatGPT Vision excels at high-level image captioning and contextual storytelling, especially for stylized or domain-specific imagery. The two approaches provided complementary insights: one grounded in traditional classification and the other in narrative interpretation.

Our analysis showed that the identified objects were generally absent from the images' generated captions and the original haunted place descriptions. While some detected objects aligned with structural elements, others were unrelated to haunted themes. This outcome highlights the limitations of off-the-shelf object detection models when applied to AI-generated or surreal images. In our results, the detected objects were often generic and disconnected from the haunted place content, which is expected given the model's training and intended use.

GeoTopic Parser

There are clear geographic patterns in the haunted places dataset. The highest number of hauntings are reported in California (739 posts), Texas (538 posts), and Ohio (410 posts). We can observe this by extracting all location mentions from the haunted place descriptions, querying the Lucene Geo Gazetteer to resolve them to real-world places, and counting how often each location appeared across the dataset. Urban legends or regional folklore might play a big role in this because California and Texas have many cultural stories.

Moreover, there are correlations between the cities with haunted places and the entities identified. This was uncovered by running a script that finds what kinds of entities are most associated with top cities. For instance, in Ada, references to Old Ada Cemetery, Egypt Valley Country Club, and Town of Honey Creek indicate that cemetery and rural landmarks are frequently associated with hauntings in that area. Similarly, Albion is tied to Albion College, suggesting that campus-related paranormal activity might be a theme in the city. In Adrian, mentions of Siena Heights University, Sand Creek, and Mei King Mansion show a clustering of ghost stories around educational institutions and local landmarks. These patterns suggest that hauntings aren't randomly distributed but tend to center around culturally significant locations within each city.

SpaCY

SpaCY[1] is an open source, industrial-strength toolkit for natural language processing in Python. For assignment two, we used the *en_core_web_trf* model[2], the largest and most accurate of SpaCY's pre-trained Named Entity Recognition taggers. We briefly explored creating a custom NER pipeline, however, this requires thousands of hand annotated haunted places entries - defeating the purpose of using an automated tool.

SpaCY successfully extracted named entities from 83.52% of the descriptions for a total of 34,044 entities, 14,921 of which were unique. "DATE", "TIME", "FAC", and "PERSON" were the most common, telling us that witnesses often have timeline and location information in their accounts. The "PERSON" entities reveal more about the apparition of a haunted place than the witnesses. For example, "Mary" was identified 64 times and "Jesus" was identified 12 times, strengthening the connection between religious iconography and haunted places we made in assignment one. Additionally, these entities reveal niche ghosts like: "Al Capone", who can be found haunting the site of the Valentine's Day Massacre and other locations along the rustbelt[3]. Or take "Ed", a man who died in an internet cafe playing "EverQuest". These story-rich entities provide important context to our dataset.

The extracted entities also revealed information about the authors of the dataset, most notably we discovered a spur of 167 entries with dates between March 2008 – June 2008. These entries begin with

[1] SpaCY
[2] en_core_web_trf
[3] ./reports/referencedPlaces/alCaponeIndicies.txt

“[Month] 2008 Update/Correction”, use “-“ as a delimiter, disprove/debunk the witness account, and use all caps for emphasis. Given their similarity in structure and style, we concluded that these entries were written by a single, opinionated author we dubbed “The Keyboard Warrior” [4]. To illustrate our warrior in action, we highlight haunted place 1368 ~ “ June 2008 Correction/ additional information: the human sacrifice story is ridiculous.” SpaCY not only identified the entities within the text, but the entities who wrote the text.

Lastly, the SpaCY entities revealed critical edge cases we missed in assignment one. For example, our previous “extract_dates” function failed to parse holidays and centuries. We updated our “classify_time_of_day” and “extract_dates” functions to account for these oversights. These changes resulted in a 3% and 2% increase in coverage across our dataset for date and time of day respectively. We compiled all changes in a detailed report[5]. In conclusion, NER provided critical insights into our haunted places, context about the authors of our dataset, and helped improve upon results from assignment one.

Tika’s Show and Tell Caption Generator

Tika’s Show and Tell Caption Generator is a neural image captioning model that uses InceptionV3 for visual feature extraction and a recurrent neural network (RNN) to generate descriptive sentences. We accessed it through the im2txt-rest-tika Docker container, which exposes the model via the /inception/v3/caption/image endpoint. When an image is submitted, the model generates multiple candidate captions, each with an associated confidence score. For each image, we used a Python script to select the “best caption,” or the one with the highest confidence score, and saved it to the Image_Caption column in the haunted_places_features_added_v2.tab file. The process was applied across 9,888 haunted place images (out of 10,992 total haunted places), resulting in one caption per image.

To evaluate the quality of these captions, we manually reviewed a random subset of 100 images and their associated captions. We categorized each caption as “Accurate” if it clearly and correctly described what was visible in the image; “Partially Accurate” if it was generally relevant but included either missing or incorrect details (for example, referencing a “house with a clock tower” when there was a house but no clocktower, or claiming a “train” was present when only tracks were visible); or “Inaccurate” if it either hallucinated entire objects or fundamentally misunderstood the scene. Out of the 100 images reviewed, 22% were categorized as Accurate, 40% as Partially Accurate, and 38% as Inaccurate. These results highlight the challenges of automated captioning, particularly when dealing with vague and visually subtle content, like abandoned buildings, cemeteries, or other haunted environments that lack clear semantic cues.

During the manual review, we also observed several notable trends. For one, many of the captions began with the phrase “a black and white photo of…,” even when the images were in color. There was also frequent misidentification of features as “clock towers,” “benches,” or “two people” when only one person appeared. We also noticed that outdoor scenes and cemeteries were especially prone to misinterpretation. To further investigate these trends, we conducted a word frequency analysis of all 9,888 image captions using CountVectorizer. The top words appearing across captions were “white” (11.73%), “black” (11.48%), “photo” (11.42%), “building” (6.19%), and “clock” (5.31%). Terms like “bench,” “train,” and “brick” also appeared frequently. These results support our qualitative observations and reveal that the model heavily leans on certain generic or repetitive phrases, likely due to limitations in the training data or biases in the image domain.

In conclusion, while Tika’s Show and Tell model enabled large-scale caption generation across our dataset, the results revealed some issues in accuracy and consistency. Despite these limitations, the captions still provide a valuable layer of metadata and help inform broader trends in the visual features associated with haunted places.

Thoughts on Tools

[4] ./reports/referencedPlaces/keyboardWarriorIndicies.txt

[5] reports/namedEntityRecognitionReport.md

[GeoTopic Parser]

Using lucene-geo-gazetteer to download allCountries.txt and build the geoIndex worked smoothly with just a few terminal commands. Also, launching the REST API with lucene-geo-gazetteer -server was simple. However, we ran into many errors with 'Invalid signature file digest' when Tika wouldn't run properly under JDK 17. Also, Installing dependencies was frustrating because it required rebuilding the virtual environment over and over again due to Python management restrictions. Moreover, figuring out which custom files needed to be added to the classpath and which ones Tika actually picked up required trial and error. It was difficult to get consistent GeoTopicParser results from the haunted places dataset since the parser missed many locations or returned NaN. Thus, we integrated spaCy NER and Gazetteer REST as a fallback solution.

[SpaCY]

SpaCY was incredibly easy to install and implement out of the box. The pre-trained pipelines were all run in a single Jupyter notebook and took 10 minutes to run through the entire dataset. Our only critique is that these pre-trained pipelines do not output confidence scores. There was a brief mention of adding this feature on GitHub[6], but as of right now, we would need to train our own model to output confidence scores.

[Tika Image Captioning]

Using Tika's Show and Tell Caption Generator was relatively easy to set up through the im2txt-rest-tika Docker container, which exposes a local REST API for image captioning. Once running, the service made it very easy to send images and receive captions in return, saving around 2,000 captions per 20 minutes. However, I needed to download all 9,888 haunted place images to my computer in order to send them through the API, which filled my disk storage and forced me to delete personal files off my computer to be able to complete the task. For this reason, managing storage and coordinating the batch processing was the most difficult part of the workflow.

[Tika Docker and InceptionV3]

We initially attempted to use Tika's inception-rest-tika Docker container but ran into several issues on Apple Silicon. The container was built for x86 systems, which made it incompatible with our setup. Although we tried modifying the Docker scripts, the service continued to launch on the wrong port and defaulted to the GeoTopicParser instead of the intended image classification endpoint. Ultimately, the container could not be adapted to work in our environment. We transitioned to running InceptionV3 directly through TensorFlow in Python. The model was easy to load and apply, but as expected, it was limited to ~1,000 ImageNet categories, and most were unrelated to haunted or supernatural themes. While we could not run the Tika Docker container with our system, implementing the same model locally allowed us to complete the object detection task with control and consistency.

Conclusion

This assignment demonstrated how large-scale content extraction and multimodal analysis tools can be integrated to uncover patterns in a dataset like Haunted Places. We combined geolocation resolution through GeoTopicParser, entity recognition with SpaCY, and image-based analysis using Tika's Show and Tell captioning model and InceptionV3 object detection, adding textual and visual metadata to the dataset. Named entity recognition revealed religious references, historical figures, and authorship patterns. Our visual analysis process added further insight by generating, captioning, and analyzing over 9,800 haunted scenes and using object detection and contextual interpretation to extract meaning from the images. Together, these tools transformed unstructured witness accounts into structured metadata and visual narratives, highlighting the paranormal accounts.

[6] Github_Discussion